

MSE, MAE, RMSE aur R^2 / Adjusted R^2

Regression Performance Metrics — Detailed Notes (Hinglish)

1. Shuruaat: Cost Function kya hota hai?

Jab hum ek Linear Regression model banate hain (jaise Experience se Salary predict karna), to model ek best-fit line banata hai. Har data point (actual value) aur line par predicted value ke beech ek gap hota hai — isko hum Error ya Residual (e) kehte hain.

- e = actual value (y) – predicted value ($\hat{y}$)

Ab is error ko measure karne ke liye hume ek 'Cost Function' chahiye — ek aisa formula jo bataye ki humara model kitna achha ya kharab predict kar raha hai. Isi cost function ko minimize karke hum model ke best parameters (θ_0, θ_1) dhoondte hain — yehi Gradient Descent ka core idea hai.

Yaad rakho: Cost Function $\rightarrow$ Performance Metric bhi hai. Isi se hum decide karte hain model achha hai ya nahi, aur isi se model train (optimize) bhi hota hai.

Convex vs Non-Convex Function (Global vs Local Minima)

Jab cost function $J(\theta_0, \theta_1)$ ko plot karte hain, to graph ek 'U' shape (bowl shape) me aata hai — isko Convex Function kehte hain. Convex function me sirf ek hi minimum point hota hai jise Global Minima kehte hain — yani sabse kam error wala point.

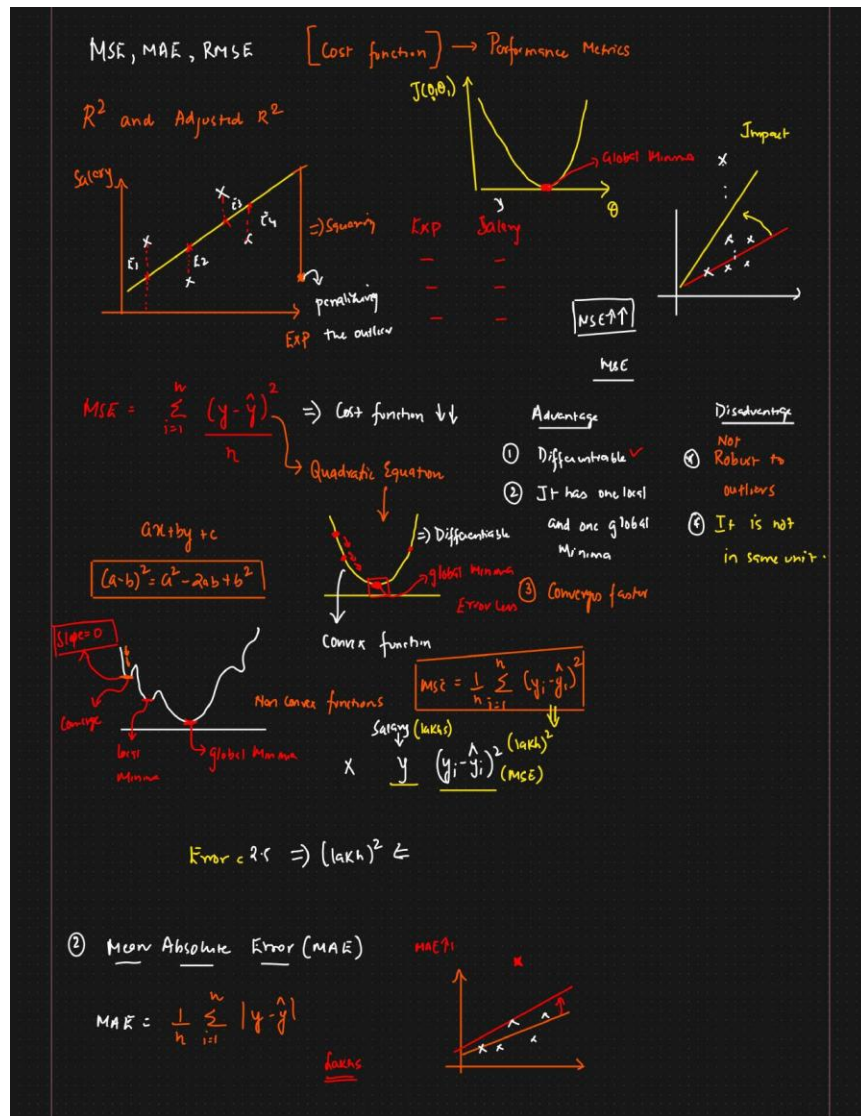
Agar function 'U' shape me nahi hai balki usme multiple ups-and-downs hain (Non-Convex Function), to usme kai Local Minima ban jaate hain — jaha slope = 0 ho jata hai lekin wo sabse best (lowest) point nahi hota. Gradient Descent kabhi kabhi in local minima me hi fas (converge) ho sakta hai, jo galat/sub-optimal solution deta hai.

- **Global Minima** — poore function ka sabse lowest point (best solution).
- **Local Minima** — apne aas-paas ka sabse lowest point, but overall best nahi.
- **Slope = 0** wahi point hota hai jaha function converge karta hai (minima pe).

R^2 aur Adjusted R^2 — Outliers ka Effect

Agar salary vs experience wale data me koi outlier (bahut door ka galat/extreme point) aa jaye, to wo line ke fitting ko bahut zyada disturb kar sakta hai. Isiliye error ko square karke (squaring) outlier ko extra penalize kiya jata hai — jitna bada error, utna zyada uska square badhega, jisse model us outlier ko zyada seriously leta hai aur usko handle karne ki koshish karta hai.

Iska seedha impact regression line ke slope aur R^2 (jo batata hai model data ko kitna achha explain kar raha hai) par padta hai — jyada outliers $\rightarrow$ jyada distorted line $\rightarrow$ R^2 ka galat estimate.



Original Notes — Page 1: Cost Function, MSE aur MAE

2. Mean Squared Error (MSE)

MSE sabse common cost function hai jo regression me use hoti hai. Ismein har data point ka error nikal ke usko square karte hain, phir sabka average (mean) le lete hain.

$$MSE = (1/n) \times \sum (y_i - \hat{y}_i)^2 \text{ [for } i = 1 \text{ to } n]$$

Notation ka matlab (term by term):

- **n** = total number of data points (kitne observations hain dataset me)
- **i** = index — 1st point se n-th point tak loop chalta hai ($i = 1, 2, 3 \dots n$)
- y_i = i-th data point ki actual (real) value — jaise actual Salary
- $\hat{y}_i$ (y -hat) = i-th data point ke liye model ki predicted value — jaise model ne predict ki Salary
- $(y_i - \hat{y}_i)$ = error / residual — actual aur predicted ke beech ka difference
- $(y_i - \hat{y}_i)^2$ = error ko square karna — isse (a) negative errors positive ban jaate hain, aur (b) bade errors ko zyada weight milta hai

- Σ (**sigma**) = summation symbol — sabhi n points ke squared errors ko add karna
- $(1/n)$ = total sum ko n se divide karke average (mean) nikalna

Example samjhein: agar y = Salary (in lakhs) hai aur error = $(y_i - \hat{y}_i)$ hai, to error ki unit bhi 'lakh' hogi. Lekin jab hum error ko square karte hain, to uski unit ban jaati hai 'lakh²' — yehi wajah hai ki MSE ki value ka unit original data ke unit se match nahi karta.

MSE kyun ek Quadratic Equation banata hai?

Formula $(a-b)^2$ ko expand karo to milta hai: $a^2 - 2ab + b^2$. Isi tarah $(y_i - \hat{y}_i)^2$ ko θ (model parameters) ke sath expand karne par ek Quadratic Equation ($ax^2 + by + c$ form) banta hai. Quadratic equation ka graph hamesha ek smooth 'U' shape (Convex Curve) banata hai — jo Differentiable hota hai (yani har point par uska ek clear, well-defined slope/derivative hota hai). Isi property ki wajah se Gradient Descent MSE ko efficiently minimize kar pata hai.

MSE ke Advantages:

- **1. Differentiable ✓** — Quadratic (convex) shape hone ki wajah se har point par derivative nikala ja sakta hai, isliye Gradient Descent smoothly kaam karta hai.
- **2. Ek hi Local aur ek hi Global Minima** — convex function hone ki wajah se confusion nahi hota ki model kaha converge karega; sirf ek best point hota hai.
- **3. Converge Fast** — smooth curve hone ki wajah se optimization algorithm jaldi best solution tak pahunch jata hai.

MSE ke Disadvantages:

- **1. Robust to Outliers nahi hai ✗** — squaring karne ki wajah se agar ek bhi bada outlier ho, uska square bahut zyada bada ho jata hai aur pura MSE us ek point ki wajah se bahut zyada badh jata hai. Isse model outlier ko fit karne ki jyada koshish karega, jo asal me sahi nahi hai.
- **2. Same unit me nahi hota ✗** — jaise upar dekha, agar y ki unit 'lakh' hai to MSE ki unit ban jaati hai 'lakh²'. Isse humein directly ye samajhna mushkil hota hai ki error 'kitne lakh ka hai' — hume ise interpret karna padta hai.

3. Mean Absolute Error (MAE)

MAE me har point ke error ko square karne ke bajaye uska Absolute Value (modulus, $|\dots|$) le lete hain — yani negative sign hata dete hain, phir average nikalte hain.

$$\text{MAE} = (1/n) \times \Sigma |y_i - \hat{y}_i| \quad [\text{for } i = 1 \text{ to } n]$$

Notation ka matlab:

- $|y_i - \hat{y}_i|$ = actual aur predicted value ke difference ka Absolute Value (modulus) — matlab sirf magnitude, sign (+/-) ignore kar diya jata hai
- $n, i, y_i, \hat{y}_i, \Sigma, (1/n)$ = same meaning jaisa MSE me tha (total points, index, actual, predicted, summation, average)

Yaha error ki unit original data jaisi hi rehti hai — jaise agar y 'lakh' me hai to MAE bhi 'lakh' me hi aata hai (kyunki hum square nahi kar rahe).

MAE ke Advantages:

- **1. Robust to Outliers ✓** — squaring nahi hone ki wajah se ek bada outlier bhi poore metric ko utna zyada distort nahi karta jitna MSE me karta hai.

- **2. Same unit me hota hai ✓** — original data (y) jis unit me hai, MAE bhi usi unit me rehta hai, isliye interpret karna aasan hota hai.

MAE ke Disadvantages:

- **1. Not Differentiable at 0 ✗** — Absolute value function $|x|$ ka graph 'V' shape ka hota hai. Iske sabse niche wale point ($x = 0$) par ek 'sharp corner/kink' hota hai — wahan slope achanak se change ho jata hai, isliye normal derivative define nahi hoti.
- → **Subgradient:** jaha function differentiable nahi hota (jaise $|x|$ ka corner point), waha ek special concept use hota hai jise 'Subgradient' kehte hain — ye us point par possible slopes ki ek range deta hai taaki optimization phir bhi chal sake, lekin isse process thoda complex ho jata hai.
- **2. Convergence usually slow / complex ✗** — sharp corner ki wajah se Gradient Descent ko optimize karne me zyada time lag sakta hai — ye ek complex optimization task ban jata hai.
- **3. Time Consuming ✗** — upar wale point ki wajah se overall training process me zyada time lagta hai.

4. Root Mean Squared Error (RMSE)

RMSE simply MSE ka Square Root hota hai. Isse MSE ki 'squared unit' wali problem solve ho jaati hai.

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\left(\frac{1}{n} \times \sum (y_i - \hat{y}_i)^2 \right)}$$

Notation ka matlab:

- $\sqrt{\text{(square root)}}$ = MSE ki value ka square root nikalna
- **Baaki sab terms** = same jaise MSE me the ($n, i, y_i, \hat{y}_i, \Sigma$)

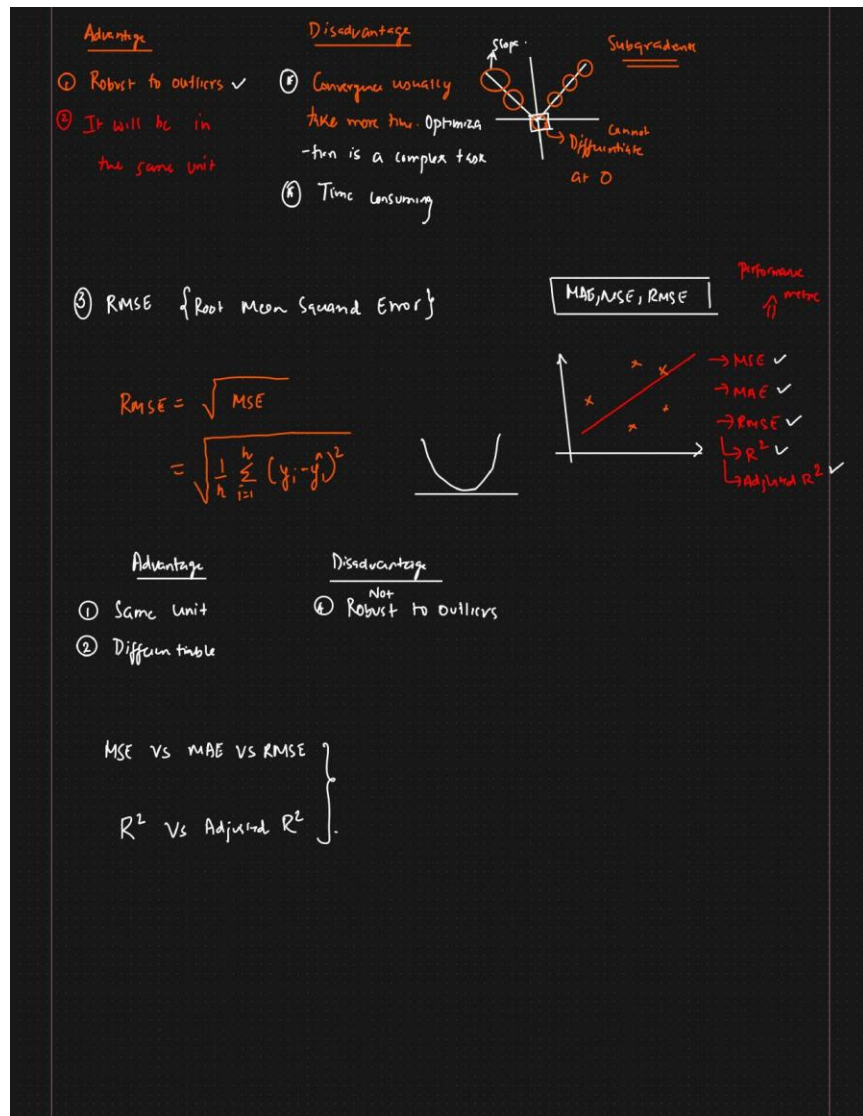
Jaise MSE me error square hoke uski unit 'lakh²' ban gayi thi, RMSE nikalne se wapas square root le liya jata hai, isliye final unit wapas 'lakh' (original unit) ban jaati hai.

RMSE ke Advantages:

- **1. Same unit ✓** — square root lene ki wajah se RMSE ki unit original data (y) jaisi hi ho jaati hai, isliye interpret karna easy hai.
- **2. Differentiable ✓** — MSE se derive hone ki wajah se ye bhi ek smooth, convex-type curve deta hai jo optimize karna aasan hota hai.

RMSE ka Disadvantage:

- **1. Not Robust to Outliers ✗** — kyunki ye MSE se hi banta hai (jisme squaring hoti hai), isliye outliers ka effect isme bhi utna hi zyada rehta hai jitna MSE me tha, sirf final unit theek ho jaati hai.



Original Notes — Page 2: MAE (Disadvantages), Subgradient, RMSE

5. Quick Comparison — MSE vs MAE vs RMSE

Metric	Formula	Robust to Outliers?	Same Unit as Data?
MSE	$(1/n)\sum(y_i - \hat{y}_i)^2$	Nahi ✗	Nahi (squared unit) ✗
MAE	$(1/n)\sum y_i - \hat{y}_i $	Haan ✓	Haan ✓
RMSE	$\sqrt{MSE}$	Nahi ✗	Haan ✓

In teeno ko ek sath kab use karte hain?

Kisi bhi regression model ki performance judge karne ke liye aksar MSE, MAE, aur RMSE — teeno ko report kiya jata hai, kyunki har ek alag angle se model ke error ko dikhata hai: MAE aasan interpretation deta hai, MSE training ke liye achha (differentiable + outlier-sensitive) hota hai, aur RMSE original unit me error dikhata hai.

Aage kya (Notes me mention hai, next topic):

- **R^2 (R-squared)** — batata hai model apne data ki variation ka kitna % explain kar pa raha hai.
- **Adjusted R^2** — R^2 ka improved version, jo extra useless features add karne par penalize karta hai (R^2 ki tarah blindly badhta nahi hai).

Notes me R^2 aur Adjusted R^2 ka topic 'aage discuss karenge' likha hua hai (last page me 'MSE vs MAE vs RMSE' aur ' R^2 vs Adjusted R^2 ' dono next-topics ke roop me note kiye gaye hain) — agar chaho to iska bhi ek detailed alag document bana sakta hoon.